



Queues in Web Applications

This article explores message queues—a technique that can be used to improve the user-perceived responsiveness of Web applications.

If you are building a Web application, it is obviously to be used by people—and you want them to be happy when they use it. When I am talking about Web applications, I don't only mean websites, but also mobile and desktop apps that make use of the Internet to fetch some data from your server, or another Web service. Your users' happiness depends on a lot of factors. Usability is one of them, of which *responsiveness* is an integral part. Your users unknowingly want to use responsive apps that don't make them wait while they complete some task, but allow users to carry on with whatever they want while the app is busy working in the background.

The problem

The problem with an app's responsiveness arises because, generally, our first attempt at building our application and handling requests has unrealistic aims. We often want to

do everything in *one request-response cycle* (Figure 1). For example, a user confirms her order on an e-commerce website; the server receives this request, and creates relevant entries in the database. It then sends a notification email to the user about the order's confirmation, and then sends the response back to the client. Now, how important is it to send the email before giving the user feedback? The email step would usually take longer, compared to other things your application does while processing this particular request. Wouldn't it greatly reduce your response time if you could somehow complete the email-sending task after the user response is sent? This way, your users don't have to wait while the server is busy doing a less important task. They can continue with whatever they wish to do, while your application takes care of the less important work at the back.